

Code-Layout, Readability and Reusability

Date	15 March 2026
Team ID	SWTID-2026-9392
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	PEP 8 / Standard format
2	Proper File Structure	Files and folders are logically organized	Yes	Separated routes, components & utils
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Self-descriptive variable naming
4	Function / Method Names	Functions are descriptively named	Yes	Action-verb prefixes used
5	Code Comments	Inline and block comments explain logic	Yes	Docstrings on key APIs
6	Modular Design	Code is split into reusable functions/modules	Yes	Gemini helper decoupled
7	No Redundant Code	Duplicate or unused code is removed	Yes	Refactored helper classes
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Try-catch blocks for LLM API calls

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Gemini Storyboard Handler	Python / Google GenAI SDK	Plot generator & scene layout builder	High

Reusable Components / Modules (Continued):

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
2	Comic Panel Grid Renderer	JavaScript / React / CSS Grid	Story canvas & export preview page	High
3	Prompt Builder Utility	Python	Character consistency & style parameters	High
4	PDF Export Generator	Python / HTML-to-PDF	Download final comic book module	Medium
5	API Error Wrapper	Python / FastAPI	All backend service endpoints	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5 / 5	Clear directory layout separating core Gemini logic, views, and assets.
Readability	5 / 5	Clear function names, type hints, and self-documenting code style.
Reusability	4.5 / 5	Core LLM prompt templates and UI card components are cleanly modular.
Documentation / Comments	4.5 / 5	Includes docstrings and inline comments on complex API handlers.
Overall Score	4.8 / 5	High-quality, maintainable codebase ready for extension.